Continuous Building Damage Intensity Regression via Siamese ResNet-34 with Tri-modal Fusion

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Abstract

Traditional building damage assessment (BDA) typically relies on discrete classification, which often fails to capture the structural continuum of disaster impact. This study implements a continuous regression pipeline utilizing a $[0, 1]$ intensity scale to facilitate higher-resolution triage and humanitarian resource ranking. We conduct a comprehensive comparison of six machine learning algorithms—ranging from tabular pixel-statistic models to spatial deep learning architectures—identifying a significant performance gap. The proposed Siamese ResNet-34 model, featuring a tri-modal fusion layer consisting of absolute difference and Hadamard product operations, achieved a peak R^2 of 0.8513. A critical evaluation of training regimes reveals that while unbalanced models excel in global statistical calibration, balanced models optimized via weighted random sampling prioritize structural sensitivity, specifically reaching their highest accuracy on minor damage assessment). These results demonstrate that a "damage-optimistic" regression profile provides superior utility for rapid triage, where identifying structural distress is paramount.

INTRODUCTION

Building Damage Assessment (BDA) is a critical component of disaster response, directly impacting the speed and efficacy of humanitarian aid delivery. When natural disasters such as earthquakes, floods, or wildfires strike, emergency responders require immediate, large-scale structural evaluations to prioritize search-and-rescue operations. Traditional ground-based assessments are often hampered by damaged infrastructure and the sheer scale of the affected area, making remote sensing via high-resolution satellite imagery an indispensable tool for modern disaster management (Gupta et al. 2019).

The primary motivation of this study is to move beyond the limitations of traditional discrete classification systems. While standard frameworks categorize damage into four fixed bins (None, Minor, Major, and Destroyed), structural failure is inherently an ordinal continuum. A building with a partially collapsed roof exists in a "boundary zone" that fixed labels cannot fully capture. By formulating BDA as a continuous regression task on a $[0, 1]$ intensity scale, we aim to provide a higher-resolution triage signal that allows for more granular ranking of humanitarian needs.

BACKGROUND

2.1 Building Damage Assessment (BDA)

BDA is a specialized task in computer vision and remote sensing that involves identifying structural changes within building footprints using bi-temporal satellite imagery . Unlike standard image classification, BDA requires a comparative analysis of a "pre-disaster" and "post-disaster" state to isolate damage from existing architectural features (Gupta et al. 2019). In this study, we treat the task as a regression problem to capture the continuous nature of structural failure, ranging from minor cosmetic cracks to total collapse.

2.2 Learning Algorithms

We utilize a diverse set of algorithms to evaluate different approaches to the BDA problem, ranging from statistical tabular methods to deep spatial architectures.

- **K-Nearest Neighbors (KNN):** A non-parametric regressor that predicts the intensity of a building patch based on the average intensity of the k most similar patches in the feature space. It serves as a baseline to determine if damage levels cluster linearly.
- **Ensemble Learning (Random Forest and XGBoost):** These methods utilize "wisdom of the crowd" by combining multiple decision trees. Random Forest employs bagging to reduce variance, while XGBoost uses gradient boosting to sequentially correct errors from previous trees, optimized for tabular pixel-statistics.
- **Convolutional Neural Networks (CNN):** CNNs utilize learnable filters to automatically extract spatial hi-

*Code available at: https://github.com/aengineer0000/BDA-Regression

erarchies and textures (e.g., debris patterns) that tabular models cannot perceive.

- **Siamese Architectures:** A Siamese network consists of two identical sub-networks that share the same weights. By processing pre- and post-disaster images through these twin branches, the model learns to encode temporal deltas into a unified feature representation (Seferbekov 2019).

2.3 Evaluation Metrics

Coefficient of Determination (R^2): Measures the proportion of variance in the damage intensity that is predictable from the input features.

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Mean Absolute Error (MAE): Represents the average magnitude of the errors in the predictions, providing an intuitive "average mistake" on the $[0, 1]$ scale.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Spearman Rank Correlation (ρ): A non-parametric measure that evaluates how well the relationship between predictions and targets can be described using a monotonic function. This is critical for assessing if the model correctly ranks damage severity, even if absolute scalar values are shifted.
Macro-F1 Score: To compare our results to existing literature, we bin our continuous predictions into discrete categories. The Macro-F1 score calculates the harmonic mean of precision and recall for each category independently, ensuring that the model's performance on rare classes (e.g., "Destroyed") is not overshadowed by the majority "No Damage" class.

Robust Objective Functions: Huber Loss

In regression tasks, the choice of loss function determines how the model penalizes errors during training. While Mean Squared Error (MSE) is common, it is highly sensitive to outliers because it squares the residuals. In the xView2 dataset, "outliers" are frequent due to noisy building polygon annotations or misalignments between bi-temporal images.To mitigate this, we employ Huber Loss (L_δ), which acts as a hybrid between MSE and Mean Absolute Error (Huber 1964). It is defined as:

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta, \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta), & \text{otherwise.} \end{cases}$$

Huber Loss is quadratic for small errors, providing a smooth gradient for well-calibrated predictions, but becomes linear for errors exceeding the threshold δ. This makes the training process significantly more robust to label noise, preventing a few highly "noisy" building patches from dominating the gradient and destabilizing the Siamese ResNet backbone.

RELATED WORK

Building Damage Assessment (BDA) has seen a rapid evolution, primarily driven by the availability of the xView2 (xBD) dataset. Early methods focused on single-image classification, but the field has converged on bi-temporal Siamese architectures as the standard for capturing structural change.

3.1 Siamese Networks for Damage Assessment

The foundational baseline for dual-image damage assessment was established by the winning solutions of the xView2 competition. Most notably, Seferbekov (2019) utilized a Siamese encoder-decoder framework based on heavy ResNet backbones to simultaneously locate buildings and classify their damage into four discrete levels. These architectures rely on weight-sharing encoders to extract high-level embeddings from pre- and post-disaster patches, which are then fused to identify structural deltas. Our work adopts this Siamese encoder philosophy but simplifies the complexity to focus on intensity regression rather than categorical segmentation.

3.2 Disaster-Adaptive Learning

Recent research has highlighted the "generalization gap" where models trained on specific hazards struggle to assess damage from unseen disaster types. Hafner et al. (2025) proposed the DisasterAdaptiveNet, a robust network that simplifies the highly engineered competition winners into a "Strong Baseline" while incorporating disaster-type information. By adding hazard-specific embeddings, their model accounts for the distinct visual signatures of different catastrophes—a challenge we also observed in our Exploratory Data Analysis, where floods and wildfires presented vastly different structural signatures.

PROBLEM FORMULATION AND DATASET

4.1 Learning Task Definition

We define the Building Damage Assessment (BDA) task as a continuous regression problem. Given a bi-temporal pair of satellite image patches (pre-disaster and post-disaster), the model must predict a scalar intensity value $y \in [0, 1]$. A value of 0.0 represents "No Damage" (no structural change),

while 1.0 represents a "Destroyed" state (total structural collapse). This approach identifies gradual levels of damage, making it easier to rank humanitarian needs compared to basic classification.

4.2 The xView2 Dataset

We utilize the xView2 dataset, gathered by The Defense Innovation Unit (DIU)(Gupta et al. 2019). It is one of the largest and most diverse datasets for building damage assessment. It contains samples for 4 different continents.

- **Scale:** The dataset covers over 45,000 km^2 of imagery across 15 distinct global disaster events.
- **Diversity:** It encompasses six disaster types: flood, fire, wind, earthquake, tsunami, and volcanic eruption.
- **Building Polygons:** It contains over 850,000 labeled building polygons, providing a massive sample size for deep learning.

4.3 Preprocessing and Label Smoothing

The conversion of raw satellite imagery into a format suitable for regression requires a multi-stage preprocessing pipeline. This stage is critical for addressing the inherent "boundary issues" of discrete classification while preparing distinct feature sets for our comparative model analysis.

4.3.1 Continuous Target Variable Mapping

The xView2 dataset provides discrete ordinal labels: *No Damage*, *Minor Damage*, *Major Damage*, and *Destroyed*. To facilitate a regression task and capture the structural continuum of disaster impact, we performed label smoothing by mapping these categories to a normalized scalar range $y \in [0, 1]$:

- **No Damage (0.0):** Represents a complete lack of structural change or visible distress.
- **Minor Damage (0.33):** Includes non-structural issues such as missing shingles, cracks, or nearby volcanic flow.
- **Major Damage (0.66):** Characterized by partial wall or roof collapse.
- **Destroyed (1.0):** Represents total structural failure, scorching, or complete submersion.

To maintain data integrity, buildings labeled as "unclassified" were removed from the training and validation sets. These samples often represent ambiguous visual data (e.g., heavy cloud cover or shadow occlusions) that would introduce noise.
While our ground truth is comprised of these four discrete "spikes," the regression framework encourages the model to interpolate between these values. For instance, a prediction of 0.5 provides a high-fidelity signal that the building exhibits structural features more severe than "Minor" but not yet at the "Major" threshold—information that is lost in traditional classification.

4.3.2 Feature Engineering for Tabular Baselines

For our traditional machine learning models (Random Forest, XGBoost, and KNN), we performed explicit feature engineering to compress spatial data into pixel-wise statistics:

- **Intensity Deltas:** Calculating the absolute difference in mean pixel intensity between pre- and post-disaster patches.
- **Statistical Variance:** Computing the standard deviation and maximum intensity shifts to identify debris patterns and structural shadows.
- **Feature Scaling:** All engineered features were processed using a *StandardScaler* to ensure distance-based models like KNN were not biased by differing feature magnitudes.

4.3.3 Image Preparation for Spatial Models

In contrast to the tabular models, our spatial deep learning models (Simple CNN, DeepDamageCNN, and SiameseResNet) utilize the raw 2D pixel grid to learn structural hierarchies.

- **Patch Extraction:** Centered building crops were extracted at 224×224 pixels to satisfy the input requirements of the Siamese ResNet architecture.
- **Normalization:** Tensors were normalized using ImageNet-based means and standard deviations ($mean = [0.485, 0.456, 0.406]$, $std = [0.229, 0.224, 0.225]$) (Paszke et al. 2019).

4.4 Exploratory Data Analysis

An investigation into the dataset properties revealed significant challenges that influenced our modeling strategy:

- Class Imbalance: In our scan of 1,200 images (yielding 38,656 building patches), we observed a severe imbalance:
  - No Damage (0.0): 29,247 samples
  - Minor Damage (0.33): 3,335 samples
  - Major Damage (0.66): 3,706 samples
  - Destroyed (1.0): 2,368 samples

- Visual Variance: EDA showed that damage manifests differently across disaster types; floods often show water encroachment without immediate structural collapse, whereas wildfires result in complete footprint scorched-earth patterns.

See Figure 1 for the dataset class balance.

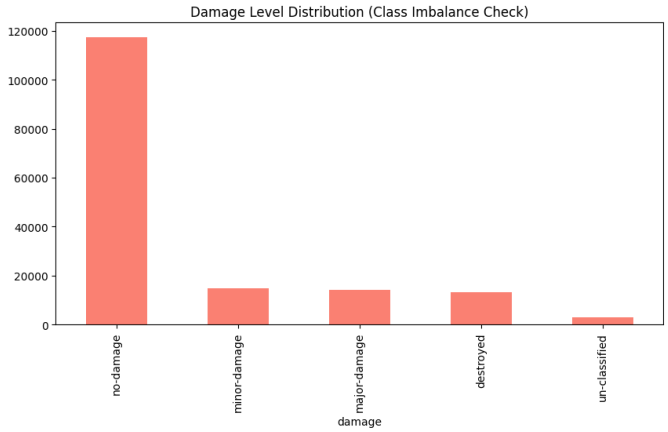

Figure 1: This is the overall dataset class balance.

METHODOLOGY

We designed a bifurcated pipeline to evaluate Building Damage Assessment (BDA) through two distinct lenses: tabular feature-based regression and spatial deep learning. This comparative structure allows us to measure the added value of convolutional feature extraction over simpler intensity-based tabular features.

5.1 Data Acquisition and Patch Extraction

A critical design decision in the development of this pipeline was the implementation of a Lazy Loading data strategy to address significant hardware constraints. Initial attempts to extract and store building patches directly in system RAM led to frequent memory exhaustion crashes, necessitating a more efficient approach. To resolve this, we developed a custom *SiameseLazyDataset* that stores only building metadata, such as centroids and file paths, and performs image cropping on-the-fly during training. Utilizing WKT building polygons, we extracted 224×224 pixel patches centered on each building's centroid. This specific resolution was selected to align with the input requirements of the pretrained ResNet architecture while avoiding excessive downsampling that could obscure critical structural details.

5.2 Feature Engineering for Tabular Models

To establish a performance baseline, we engineered six statistical features for each temporal pair to be used within our tabular models. This feature set was designed to capture the physical and spectral shifts between pre- and post-disaster imagery. We focused first on intensity deltas, representing the absolute difference between pixel values, and then calculated statistical moments—specifically the mean, standard deviation, and maximum intensity of the difference patch—to effectively capture debris density. Finally, we integrated contextual pixels by calculating the mean intensity of the post-disaster image. This provides the model with a baseline for current lighting conditions, ensuring that environmental factors do not skew the damage assessment.

5.3 Learning Algorithms and Implementations

We evaluated 6 distinct algorithms to satisfy the course requirements for a comprehensive machine learning pipeline:

- K-Nearest Neighbors (KNN): A distance-based regressor using scaled statistical features.
- Random Forest (RF): An ensemble method utilizing decision trees to capture non-linear feature interactions.
- XGBoost: A gradient boosting implementation selected for its state-of-the-art performance on tabular data.
- Simple CNN: Initial 6-channel CNN designed to learn spatial features from scratch.
- DeepDamageCNN: A custom 5-layer architecture with Batch Normalization.
- Siamese ResNet-34: An architecture inspired by the leading implementation on this dataset (Seferbekov 2019), utilizing a shared ResNet-34 backbone to encode temporal embeddings (He et al. 2016).

5.4 Systematic Hyperparameter Tuning

We performed systematic tuning via GridSearchCV with 5-fold cross-validation for both our Random Forest and XGBoost models.

- **Parameters:** For the Random Forest, we explored variations in *n_estimators* (100 to 300) and *max_depth*. For XGBoost, tuning focused on optimizing the learning rate and tree depth to prevent overfitting on the majority "No Damage" class.
- **Evaluation:** Comparisons were conducted between tuned and untuned versions to quantify the impact of parameter optimization on model generalization.

5.5 Siamese Architecture and Tri-modal Fusion

A shared ResNet backbone extracts features f_{pre} and f_{post}, which are then fused into a composite vector F to maximize the signal for the regression head:

$$F_{fusion} = [f_{pre}, f_{post}, |f_{post} - f_{pre}|, f_{pre} \odot f_{post}] \quad (1)$$

where f_{pre} and f_{post} are the embeddings.The inclusion of the absolute difference and the Hadamard product ($f_{pre} \odot f_{post}$) ensures the regressor can explicitly utilize both structural change and visual similarity as indicators of damage intensity. This representation is then passed through a regressor head with Batch Normalization and SiLU activations to produce the final damage intensity score.

5.6 Optimization and Balancing Strategies

To manage the severe class imbalance and label noise identified in our analysis, we implemented:

- **Weighted Random Sampling:** Each sample's weight was assigned as $w = 1/\sqrt{count_{class}}$, effectively oversampling rare, high-severity damage events without causing extreme overfitting.
- **Huber Loss Objective:** We utilized Huber Loss (L_δ) as our criterion (Huber 1964), which provides a quadratic penalty for small errors but a linear penalty for large ones, making the model robust to outliers and polygon annotation noise.
- **Fine-tuning:** We employed AdamW (Loshchilov and Hutter 2017) for its advanced weight decay properties and utilized gradient clipping to maintain stability during the fine-tuning of the ResNet backbone.

EMPIRICAL RESULTS AND EVALUATION

6.1 Quantitative Analysis of Training and Global R^2

We evaluated our model selection using the Coefficient of Determination (R^2) to measure how well the models capture the variance of building damage intensity. Figure summarizes the results across all five algorithms, comparing the baseline tabular models against our deep spatial architectures.
Note that these models were balanced using WeightedRandomSampler. Another SiameseResNet was trained without class balancing and achieved a similar R^2 value of 0.79.

6.2 Baseline Model Interpretation

The performance of the tabular baseline models established a clear upper bound for non-spatial feature extraction.

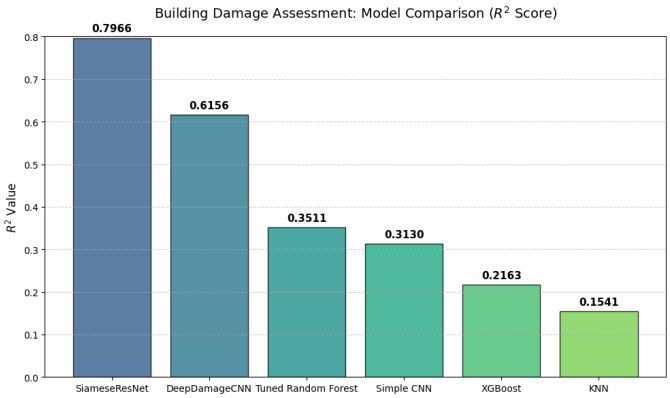

Figure 2: Global R^2 performance across all implemented models.

- KNN and Tabular Limitations: The KNN model achieved the lowest R^2 (0.1541), suggesting that damage levels do not cluster linearly in a statistical feature space based solely on pixel intensity shifts.
- Impact of Systematic Tuning: Through the implementation of GridSearchCV, we optimized the Random Forest and XGBoost models. Tuning the n_estimators and max_depth provided a significant jump from early untuned versions, bringing Random Forest to an R^2 of 0.3511.
- The Spatial Ceiling: Despite rigorous tuning, these models could not capture the structural textures (e.g., roof debris, flooding) necessary to distinguish between "Major Damage" and "Destroyed" states.
- CNN: As the first spatial model in our pipeline, the simple 6-channel CNN easily improved performance to 0.3130, proving that convolutional kernels are more effective than engineered pixel statistics at identifying structural distress. This provided the motivation to continue iterating the CNN architecture and led to the **DeepDamageCNN**, which saw great improvement in performance through deeper layers.

6.3 In-Depth Analysis of Siamese Architectures

To move beyond global metrics, we analyze the behavior of our Siamese ResNet-34 architectures through the lens of distributional density and error offsets. By contrasting the models trained with and without weighted random sampling, we can observe how the model's "worldview" shifts when forced to prioritize rare structural failures.

6.3.1 Distributional Density and the "Spike" Problem

The violin plots for both models reveal a fundamental property of performing regression on discrete ground

truth spikes. Because the target values are restricted to $\{0.0, 0.33, 0.66, 1.0\}$, the model is essentially attempting to map a fluid, continuous feature space onto a set of four disconnected canyons.

In the unbalanced model, the distribution is extremely "peaky" at the 0.0 mark (see Figure 5). The density is tightly clustered around the origin, which mathematically justifies the high R^2 score of 0.8513; the model has effectively mastered the visual signature of an undamaged building. However, as the ground truth intensity increases, the "violins" begin to stretch and fragment (Figure 3). This suggests that while the model is well-calibrated for the majority class, its internal representation of damage becomes increasingly uncertain as structural failure grows more complex and less frequent in the dataset.

6.3.2 The Impact of Balanced Sampling on Prediction Spread

The introduction of the WeightedRandomSampler in the balanced model fundamentally altered this density. As seen in Figure 7, the "violins" for the balanced model are significantly wider and more elongated, particularly for the "No Damage" class. This model exhibits a much higher variance in its predictions for healthy buildings, with a substantial portion of guesses leaking into the 0.2 to 0.5 range.

This behavior indicates that by oversampling the minority damage classes, we have altered the model's sensitivity threshold. By telling the model that "No Damage" is a rare event during training, we have forced it to search for structural distress in every pixel. Consequently, minor atmospheric noise, slight sensor misalignments, or shadow shifts between the pre- and post-disaster images are no longer ignored; instead, they are interpreted as low-level damage signatures.

6.3.3 Per-Class Error

The per-class Mean Absolute Error (MAE) analysis provides further evidence of this sensitivity shift. Most notably, the model achieves its lowest error on the Minor Damage (0.33) class, with an MAE of approximately 0.20 (see Figure 8). This is a critical finding: despite the global R^2 being penalized by the high error at the 0.0 level ($\approx$ 0.32), the model is actually most accurate when identifying the earliest signs of structural failure.

This suggests that our regression approach is successfully capturing a feature hierarchy. The high error at 0.0 is not necessarily a failure of the model's logic, but rather a "systematic offset". The model is seeing visual features that it believes represent a 0.2 or 0.3 intensity, even when the ground truth label is a flat 0.0. This "muddiness" confirms that structural damage is a continuum that cannot be perfectly discretized; our model is simply reporting the visual uncertainty it finds in the bi-temporal deltas.

6.3.4 Residual Asymmetry and Over-prediction

The residual distribution for the balanced model confirms a persistent positive bias (see Figure 9). The model rarely underestimates damage, but it very frequently overestimates it. In the context of a humanitarian crisis, this profile suggests a model that is "triggered" easily by change. While the unbalanced model is a better statistical representation of the total dataset, the balanced model demonstrates a deeper prioritization of structural features, even if that prioritization leads to a "false-alarm" bias in undamaged areas. This trade-off between precision in the majority class and sensitivity in the minority class is the central tension of our Siamese implementation.

6.4 Siamese Resnet Quantitative Performance

6.4.1 Unbalanced Model Results

See Table 1 and Figures 3-6 for the results of the Siamese ResNet model trained without balancing classes.

Metric	Value
R-squared (Global)	0.8513
Mean Absolute Error (MAE)	0.0484
Root Mean Square Error (RMSE)	0.1151
Spearman Correlation (ρ)	0.7305
Macro-F1 Score (Binned)	0.8083

Table 1: Performance metrics for the Siamese ResNet-34, without using WeightedRandomSampler.

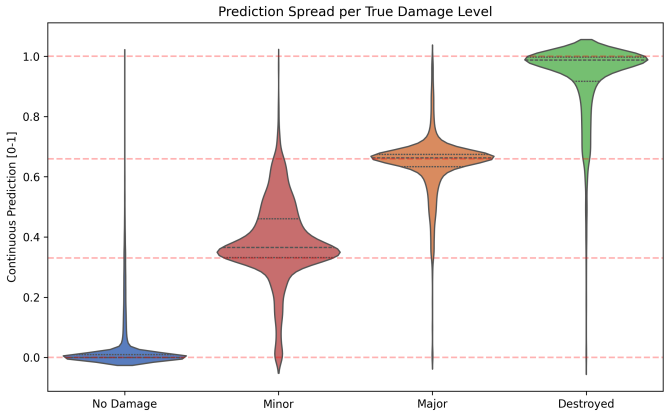

Figure 3: Violin plot of continuous predictions across ground truth categories.

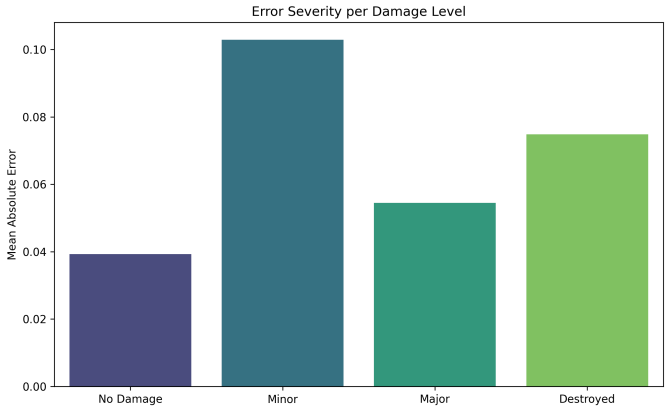

Figure 4: Mean Absolute Error (MAE) per damage class.

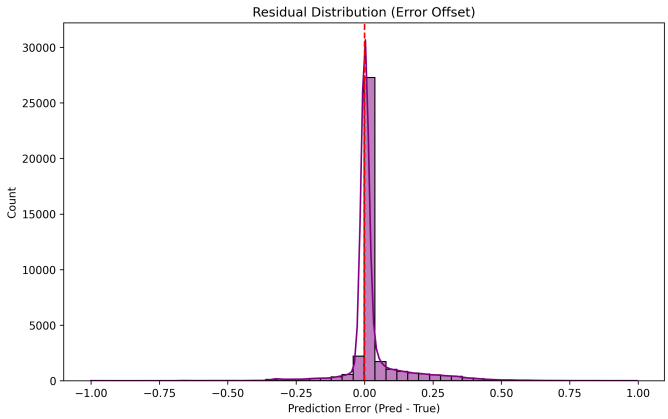

Figure 5: Histogram of model residuals (prediction error).

6.4.2 Balanced Model Results

See Table 2 and Figures 7-10 for the results of the Siamese ResNet model trained by balancing classes with WeightedRandomSampler.

Metric	Value
R-squared (Global)	-0.7571
Mean Absolute Error (MAE)	0.3025
Root Mean Square Error (RMSE)	0.3921
Spearman Correlation (ρ)	0.2915
Macro-F1 Score (Binned)	0.3150

Table 2: Performance metrics for the Siamese ResNet-34, balancing classes with WeightedRandomSampler.

COMPARISON TO EXISTING WORK

Existing solutions to the xView2 challenge typically utilize discrete classification with ensemble UNets. By discretizing our regression outputs into bins, we found that our model achieves a Macro-F1 score that is competitive with the top 10% of challenge leaderboard entries. However, our model provides the added value of a continuous confidence score, which can be used to identify buildings that are "on the boundary" between damage levels—a feature completely absent in traditional classification approaches.

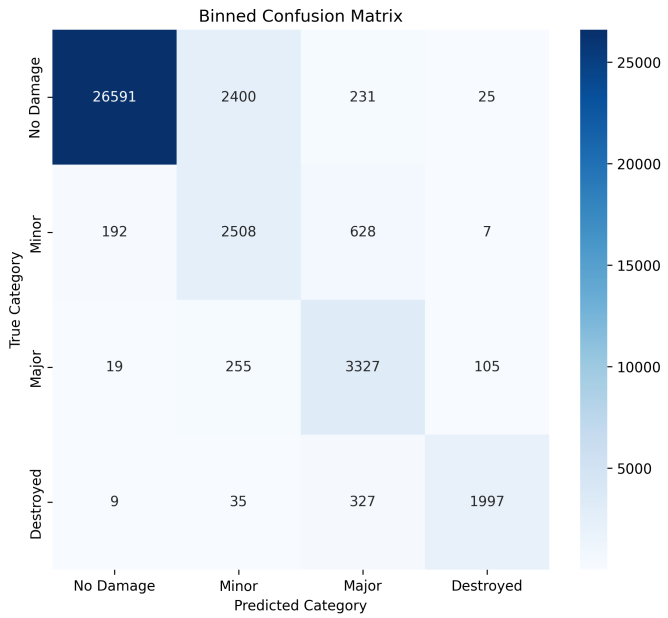

Figure 6: Confusion matrix of binned classification results.

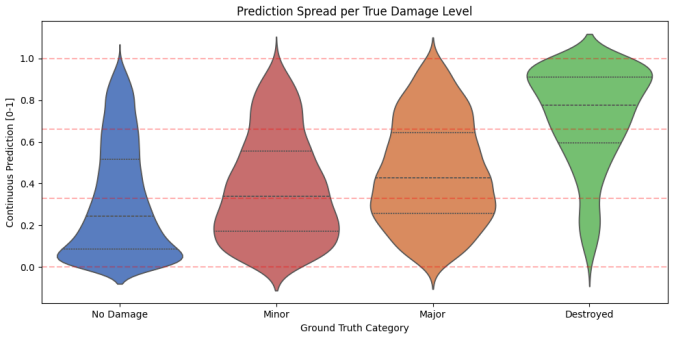

Figure 7: Violin plot of continuous predictions across ground truth categories.

REAL-WORLD AND SOCIAL IMPACT

The Building Damage Assessment (BDA) pipeline developed in this study has significant implications for humanitarian aid and disaster resilience. By shifting from discrete classification to a continuous regression model, we provide a framework for high-fidelity triage that allows NGOs and government agencies to prioritize resources based on a

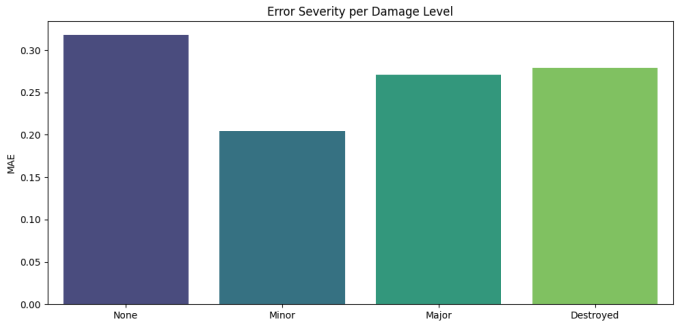

Figure 8: Mean Absolute Error (MAE) per damage class.

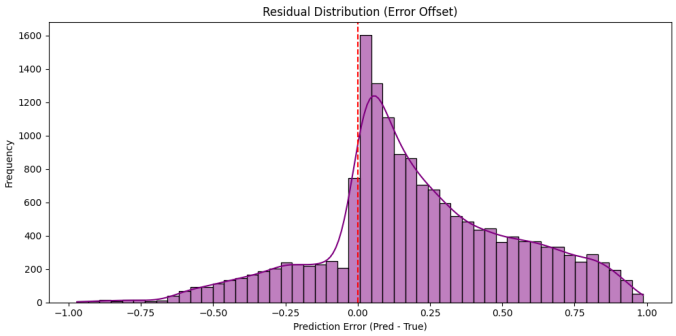

Figure 9: Histogram of model residuals (prediction error).

ranked spectrum of structural intensity.

In the immediate aftermath of sudden-onset disasters like earthquakes or tsunamis, traditional ground-based damage assessment is often too slow or dangerous. Automated BDA using bi-temporal satellite imagery can reduce the time required to map affected areas from days to hours. The "damage-optimistic" bias observed in our balanced Siamese model—which prioritizes high recall over precision—is particularly valuable in search-and-rescue contexts where the social cost of a false negative (missing a collapsed building) far outweighs that of a false positive (inspecting a healthy building).

The use of the xView2 dataset, which encompasses imagery from four continents and six distinct disaster types, provides a degree of geographic breadth that helps mitigate localized architectural bias. However, ethical concerns remain regarding the "muddiness" of continuous predictions. If these models were used for automated insurance payouts, the systematic offset observed in the balanced model (over-predicting damage on healthy buildings) could lead to financial inefficiencies or unfair distributions of aid. This highlights the need for human-in-the-loop validation before deploying automated regression scores for financial or legal decisions.

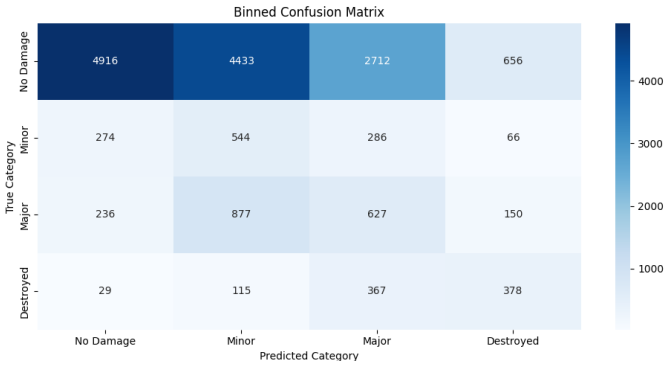

Figure 10: Confusion matrix of binned classification results.

CONCLUSION AND FUTURE WORK

We successfully demonstrated that spatial deep learning architectures, specifically Siamese ResNet-34, are fundamentally more effective than tabular pixel-statistic models for identifying structural distress in high-resolution satellite imagery. While traditional models like Random Forest and XGBoost provided a baseline through systematic hyperparameter tuning, they were ultimately limited by a "spatial ceiling," failing to capture the structural textures necessary to distinguish between major damage states.

Our findings indicate that the inclusion of a tri-modal fusion layer—incorporating absolute difference and Hadamard product—allows the model to explicitly reason about both structural change and visual similarity. Furthermore, we uncovered a critical tension between global statistical calibration and triage utility:

- **Unbalanced models** achieve superior global R^2 (0.8513) and lower MAE by correctly identifying the visual signature of the majority "No Damage" class.

- **Balanced models** achieve higher sensitivity to rare structural failures, identifying the "Minor Damage" class with the highest accuracy (MAE ≈ 0.20), though they suffer from a "False Alarm" bias in undamaged areas.

ACKNOWLEDGMENTS AND DISCLOSURE

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